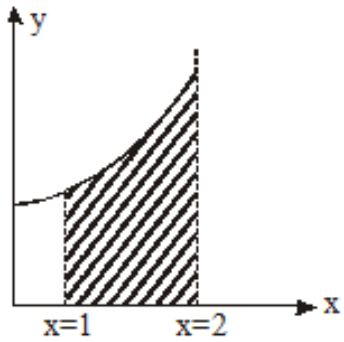
PHYSICS

1) $\int_0^{\infty} x^{-1/2}dx$

(1) 0
(2) 1
(3) 2
(4) ∞

2)

Equation of following curve is $y = 3x^2 + 1$ then area under curve between x = 1 to x = 2 is :



(1) 8 units
(2) 16 units
(3) 4 units
(4) 1 unit

3)

Value of $\int_0^1 (3x^2 - 4x + 1)dx$ is -

(1) 0
(2) 1
(3) 2
(4) 3

4) $\int(\sin x + \cos x)dx$

(1) –cosx + sinx + c

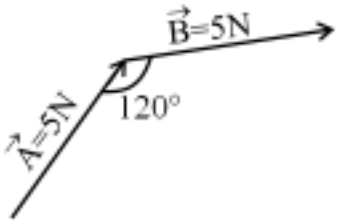
(2) sin2x + c

(3) cosx – sinx + c

(4) Zero

5)

Resultant of $\vec{A}$ and $\vec{B}$ is :-



(1) 5

(2) $5\sqrt{3}$

(3) $\frac{5}{\sqrt{3}}$

(4) 10

6) $\int_{\pi/6}^{\pi/2} \sin x \, dx$ is equal to

(1) $\frac{1}{2}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{2}$

(4) 0

7) $\int (4\cos t + t^2)dt$ is equal to -

(1) $-4\sin t + \frac{t^3}{3} + C$

(2) $-4\sin t + t^2 + C$

(3) $4\sin t + \frac{t^3}{3} + C$

(4) $4\sin t + 2t^3 + C$

8) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 dx = ?$

(1) $\frac{x^2}{2} + \log_e x - 2x + c$

(2) $\frac{x^2}{2} + \log_e x + 2x + c$

(3) $\frac{x^2}{2} - \log_e x + c$

(4) $\frac{x^2}{2} - \log_e 2x + c$

9) $\int_{R_1}^{R_2} \frac{GMm}{r^2} dr =$

(1) $GMm\left[\frac{1}{R_2} - \frac{1}{R_1}\right]$

(2) $GMm\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

(3) $GMm\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$

(4) $-GMm\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$

10) Find the sum of a series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...\infty$

(1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{5}{2}$

(4) $\frac{13}{9}$

11) $\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \frac{1}{625} + ...\infty = ?$

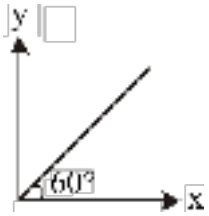
(1) $\frac{1}{4}$

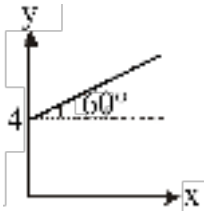
(2) $\frac{5}{4}$

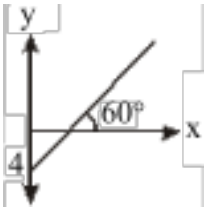
(3) 1

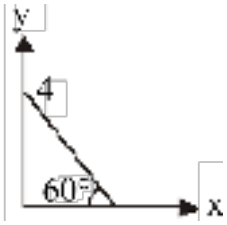
(4) Can't be determined

12) $y = \sqrt{3}x - 4$ then graph is-

(1) 

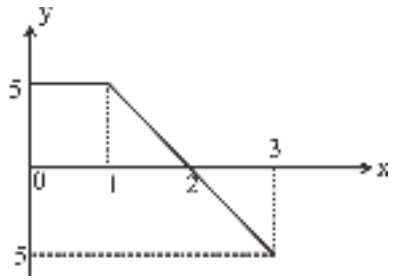
(2) 

(3) 

(4) 

13)

Find the value of $\int_0^3 y\,dx$



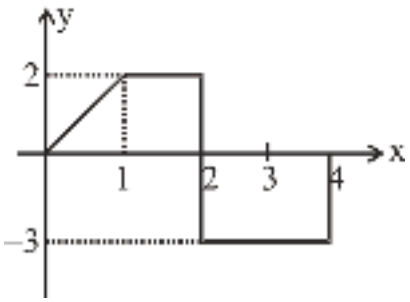
(1) 5

(2) 6

(3) 3

(4) 2

14) Area under y – x curve from x = 0 to x = 4 is equal to :-



(1) 3

(2) –3

(3) –6

(4) 9

15)

The value of $\int_0^1 9x^8 dx + \int_0^{\pi/2} \cos x \, dx$

(1) 1

(2) 3

(3) 4

(4) 2

16)

If acceleration of the particle as a function of time is given as a = 2(t – 1) m/s^2 then the value of average acceleration in the interval of zero to 2 sec will be

(1) zero

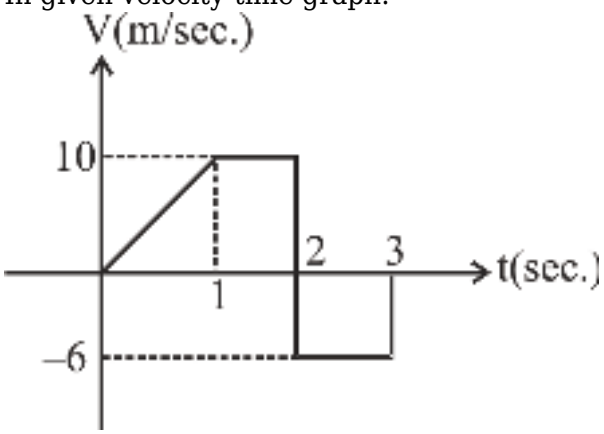
(2) 2 m/s^2

(3) –2 m/s^2

(4) None of these

17)

In given velocity-time graph:

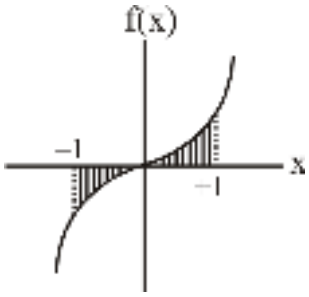


Find displacement of particle from t = 0 to t = 3 sec.

(1) 9 m
(2) 24 m
(3) 14 m
(4) 26 m

18)

If graph of a function $f(x) = x^3$ is as shown. Find the total shaded area of the graph.



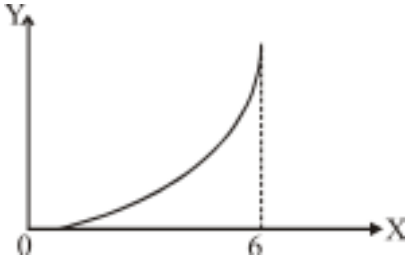
(1) 1
(2) 0
(3) $\frac{1}{4}$
(4) 2

19) $\int_{-1}^{1} e^x dx$

(1) $\frac{e^2+1}{e}$
(2) $\frac{e^2-1}{e}$
(3) $\frac{e-1}{e}$
(4) $\frac{e+1}{e}$

20)

Fig. shows the curve $y = x^2$. Find the area of the part between x = 0 and x = 6.



(1) 72

(2) 36

(3) 18

(4) 9

21)

Area bounded by y = sinx with x-axis where x varies from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ is

(1) 2

(2) 1

(3) 0

(4) $\frac{1}{2}$

22) Area bounded between the straight line y = 2x – 3 and x-axis from x = 1 to x = 4 is :-

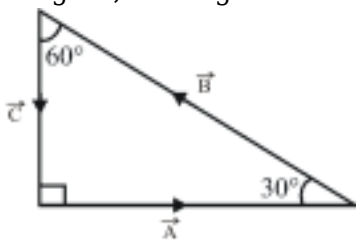
(1) 6 units

(2) 2 units

(3) 4 units

(4) 16 units

23) In the triangle shown in figure, the angle between vectors $\vec{A}$ and $\vec{B}$ is :-



(1) 60°

(2) 120°

(3) 150°

(4) 45°

24) The vector $\vec{P}$ makes 120° with the x-axis and the vector $\vec{Q}$ makes 30° with the x-axis. What is magnitude their resultant ?

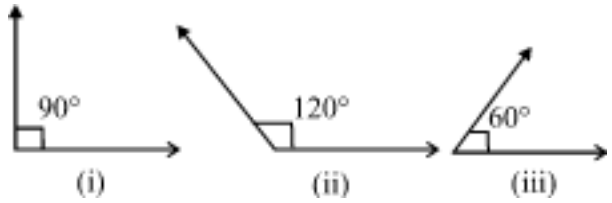
(1) P + Q

(2) P – Q

(3) $\sqrt{P^2+Q^2}$

(4) $\sqrt{P^2-Q^2}$

25) Two force vectors of same magnitude are arranged in the following manner :-



Magnitude of resultant force is maximum for :-

(1) I
(2) II
(3) III
(4) II and III

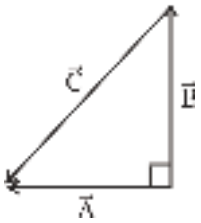
26) If $|\vec{A}| = 2$ and $|\vec{B}| = \sqrt{3}$ and angle between $\vec{A}$ and $\vec{B}$ is 30°, then resultant will be :

(1) $\sqrt{13}$
(2) $\sqrt{19}$
(3) $\sqrt{12}$
(4) $\sqrt{7}$

27) If a unit vector is represented by $0.5\hat{i} + 0.8\hat{j} + c\hat{k}$, then the value of 'c' is :-

(1) 1
(2) $\sqrt{0.11}$
(3) $\sqrt{0.01}$
(4) $\sqrt{0.39}$

28) In a triangle, shown in figure :-



(1) $\vec{A} + \vec{B} = \vec{C}$
(2) $\vec{A} + \vec{B} + \vec{C} = \vec{0}$
(3) $\vec{B} + \vec{C} - \vec{A} = \vec{0}$
(4) $\vec{A} + \vec{C} = \vec{B}$

29) Two equal vectors have a resultant equal to either of them. The angle between them is:-

(1) 60°
(2) 90°
(3) 100°
(4) 120°

30) If $\hat{A}$ is a unit vector in the direction of the vector $\vec{A}$, then :-

(1) $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$

(2) $\hat{A} = \vec{A}|\vec{A}|$

(3) $\hat{A} = \frac{|\vec{A}|}{\vec{A}}$

(4) $\hat{A} = |\vec{A}|$

31) If three vectors A, B and C are 12, 5 and 13 in magnitude such that $\vec{C} = \vec{A} + \vec{B}$, then the angle between A and B is

(1) 60°

(2) 90°

(3) 120°

(4) None

32) Out of following which resultant not possible by 10 N & 15 N

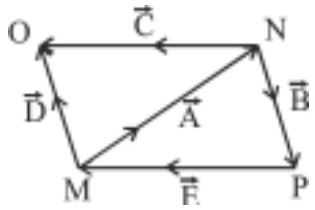
(1) 6N

(2) 24N

(3) 2N

(4) 20 N

33) From the fig., the correct relation is :-



(1) $\vec{A} + \vec{B} + \vec{E} = \vec{0}$

(2) $\vec{C} - \vec{D} = -\vec{A}$

(3) $\vec{B} + \vec{E} - \vec{C} = -\vec{D}$

(4) All of the above

34) Which of the sets given below may represent. The magnitude of three vectors adding to zero ?

(1) 2,4,8

(2) 4,8,16

(3) 1,2,1

(4) 0.5,1,2

35) Which of the following set of forces can produce zero resultant :-

(1) 20 N, 30 N, & 60 N

(2) 20 N, 40 N, & 60 N

(3) 10 N, 20 N & 40 N

(4) 10 N, 10 N & 30 N

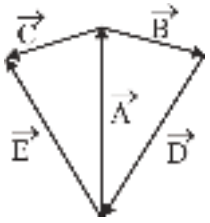
36) Two equal forces are acting at a point with an angle of 60° between them. If the resultant force is equal to $40\sqrt{3}N$, the magnitude of each force is :-

(1) 40N

(2) 20N

(3) 80N

(4) 30N

37) In figure, $\vec{D}$ equals 

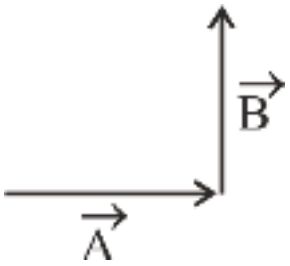
(1) $\vec{A}$

(2) $\vec{B}$

(3) $\vec{A}+\vec{B}$

(4) $-\left(\vec{A}+\vec{B}\right)$

38) Vectors $\vec{A}$ and $\vec{B}$ are shown in figure, Then diagram of is $\vec{B}-\vec{A}$:



(1) 

(2) 

(3) 

(4) None of these

39) If $\left|\vec{A}-\vec{B}\right|=\left|\vec{A}\right|=\left|\vec{B}\right|$, then angle between $\vec{A}$ and $\vec{B}$ is :-

(1) 60°

(2) 0°

(3) 120°

(4) 90°

40) If vectors $\vec{A}$ and $\vec{B}$ have an angle θ between, them, then value of $\left|\hat{A}-\hat{B}\right|$ will be :-

(1) $2\cos\frac{\theta}{2}$

(2) $2\tan\frac{\theta}{2}$

(3) $2\sin\frac{\theta}{2}$

(4) None of these

41) If $\left|\hat{a}-\hat{b}\right|=1$ then angle between $\hat{a}\&\hat{b}$ is :

(1) 0°

(2) 60°

(3) 90°

(4) 120°

42) Two force F_1 and F_2 are acting on a body. One force is double that of the other force and the resultant is equal to the greater force. Then the angle between the two forces is :-

(1) $\cos^{-1}(1/2)$

(2) $\cos^{-1}(-1/2)$

(3) $\cos^{-1}(-1/4)$

(4) $\cos^{-1}(1/4)$

43) The maximum and minimum resultant of two forces acting at a point are 10N and 6N respectively. If each force is increased by 3N, find the resultant of new forces when acting at a point at an angle of 90° with each-other :-

(1) $\sqrt{146}\,N$

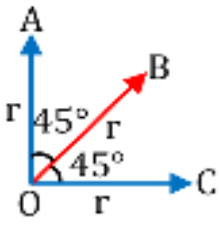
(2) 11N

(3) $\sqrt{70}\,N$

(4) 8N

44) Find the resultant of the three VECTORS $\overrightarrow{OA}$, $\overrightarrow{OB}$, and $\overrightarrow{OC}$ each of magnitude r as shown in

figure?



(1) $r(1+\sqrt{2})$

(2) $\sqrt{2}r$

(3) $2\sqrt{2}r$

(4) $r(2+\sqrt{2})$

45) Two vectors with magnitude in the ratio of 3 : 5 act at an angle of 60° and have the resultant 35. Find the magnitude of these

(1) 15, 25

(2) 15, 9

(3) 75, 125

(4) 20, 20

CHEMISTRY

1) Radius of 3rd orbit of Li^{+2} is :-

(1) 0.529 Å

(2) 1.58 Å

(3) 2.67 Å

(4) 1.2 Å

2) For Li^{+2}, $r_2 : r_5$ will be :-

(1) 9 : 25

(2) 4 : 25

(3) 25 : 4

(4) 25 : 9

3) If radius of second Bohr's orbit of He^{+} ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{2+} ?

(1) 476.1 pm

(2) 158.7 pm

(3) 52.9 pm

(4) 761.1 pm

4)

The ratio of the radii of two Bohr orbit's of H-atoms is 4 : 1 what would be their nomenclature:-

(1) K & L
(2) L & K
(3) N & L
(4) 2 & 3 both

5) Velocity of electron in 2nd orbit of He^{+} ion will be :-

(1) 4.36×10^{6} m/s
(2) 2.18×10^{6} m/s
(3) 2.18×10^{2} m/s
(4) 9.72×10^{8} m/s

6) What is the ratio of velocity of electron present in 3rd orbit of H atom and 2nd orbit of Li^{+2} ion ?

(1) 9 : 2
(2) 2 : 9
(3) 1 : 3
(4) 3 : 1

7)

What is the ratio of time periods (T_1/T_2) in first orbit to second orbit of H-atom?

(1) $\frac{4}{1}$
(2) $\frac{1}{8}$
(3) $\frac{1}{2}$
(4) $\frac{8}{1}$

8) Ratio of frequency of revolution of electron in the 3rd orbit of He^{+} and second orbit of hydrogen is :-

(1) $\frac{32}{27}$
(2) $\frac{27}{32}$
(3) $\frac{1}{54}$
(4) $\frac{27}{2}$

9) If electron falls from n = 3 to n = 2 in H-atom then emitted energy is :-

(1) 10.2
(2) 12.09
(3) 12.75
(4) 1.89 eV

10) When an electron jumps from L to K shell

(1) Energy is released
(2) Energy is absorbed
(3) Energy is neither absorbed not released
(4) Energy may be absorbed or released

11) Ist shell energy of He^{+} is –54.4 eV. Then energy of its 2nd shell is :-

(1) –54.4 eV
(2) –13.6 eV
(3) –27.2 eV
(4) +27.2 eV

12) Choose correct relation on the basis of Bohr theory:-

(1) Velocity of electron $\propto \frac{1}{n^2}$
(2) Frequency of revolution $\propto \frac{1}{n^3}$
(3) Radius of orbit $\propto n^2 \cdot Z$
(4) Energy of electron $\propto \frac{n^2}{Z^2}$

13) How much energy would be required by an electron while moving from ground state to 3rd excited state of He^{+} ion

(1) 40.8 eV
(2) 10.2 eV
(3) 51 eV
(4) 48.35 eV

14) On increasing radius (r) which of the following increases according to Bohr's model.

(1) Total energy
(2) Potential energy
(3) Both (1) and (2)
(4) Kinetic energy

15)

What is the shortest wavelength line in the Paschen series of Li^{2+} ion?

(1) $\frac{R}{9}$

(2) $\frac{9}{R}$

(3) $\frac{1}{R}$

(4) $\frac{9R}{4}$

16)

If the potential energy (PE) of electron in hydrogen is –3.02eV then in which of the following excited level is electron present :-

(1) 1st

(2) 2nd

(3) 3rd

(4) 4th

17) Which of the following relation is correct for any H like species :-

(1) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$

(2) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$

(3) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$

(4) $(E_2 - E_1) = 1/4\ (E_3 - E_2) = 1/9\ (E_4 - E_3)$

18) Calculate the potential energy of electron in 3rd orbit of H.

(1) –1.5 eV

(2) –3.02 eV

(3) –4.5 eV

(4) –2.25 eV

19) Match the following (for Z = 1) :-

(P)	$E_2 - E_1$	(i)	1.89 eV
(Q)	$E_3 - E_2$	(ii)	12.09 eV
(R)	$E_4 - E_2$	(iii)	10.2 eV
(S)	$E_3 - E_1$	(iv)	2.55 eV

(1) P-(iii), Q-(ii), R-(iv), S-(i)

(2) P-(ii), Q-(i), R-(iii), S-(iv)

(3) P-(iii), Q-(i), R-(iv), S-(ii)

(4) P-(ii), Q-(i), R-(iv), S-(iii)

20) What is the first separation energy for electron present in Be^{+3} ion ?

(1) 13.6 eV
(2) 27.2 eV
(3) 40.8 eV
(4) 54.4 eV

21)

According to Bohr theory, the radius (r) and velocity (v) of an electron vary with the increasing orbit number 'n' as :-

(1) r increases, v decreases
(2) r and v both increases
(3) r & v both decrease
(4) r decreases, v increases

22) When electron of a hydrogen atom jumps from n = 4 to n = 2 state, number of possible spectral lines emitted is :-

(1) 15
(2) 6
(3) 3
(4) 4

23) When electron jumps from 5th excited state to 2nd orbit (second orbit) in single step in H-atom then spectral line belongs to :-

(1) Lyman series
(2) Balmer series
(3) Humphrey series
(4) Pfund series

24) Which of the following electron transition will release a photon of largest wavelength in a hydrogen atom :-

(1) n = 1 to n = 2
(2) n = 4 to n = 1
(3) n = 2 to n = 3
(4) n = 2 to n = 1

25) In which of the following transition quantam of maximum frequency is emitted.

(1) n = 1 to n = ∞
(2) n = 2 to n = 1
(3) n = 3 to n = 1
(4) n = 3 to n = 2

26) In H spectrum e^- moves from 3rd orbit to 1st orbit in multistep then total no. of lines in Infrared region is :

(1) 0
(2) 1
(3) 3
(4) 4

27) In spectral series of hydrogen, the series which does not come in infrared region is:-

(1) Pfund series
(2) Brackett series
(3) Paschen series
(4) Lyman series

28) In Balmer series of hydrogen atom spectrum which electronic transition causes third line?

(1) Fifth Bohr orbit to second Bohr orbit
(2) Fifth Bohr orbit to first Bohr orbit
(3) Fourth Bohr orbit to second Bohr orbit
(4) Fourth Bohr orbit to first Bohr orbit

29) A certain electronic transition from an excited state to ground state of the H atoms sample in one or more step gives rise to three lines in the ultra violet region of the spectrum. How many lines does this transition produce in the infrared region of the spectrum :–

(1) 1
(2) 2
(3) 3
(4) 4

30) The number of possible lines of Paschen series when electron jumps from 8th orbit to ground state (in hydrogen like atom) is :-

(1) 2
(2) 5
(3) 4
(4) 3

31) **Assertion :-** Emitted radiations will fall in visible range when an electron jump from n = 4 to n = 2 in H - atom.
Reason :- Lyman series radiation belongs to visible range for H-atom.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

32) Wavelength of last line of paschen series in H spectrum is :

(1) $\frac{9}{R}$

(2) $\frac{R}{9}$

(3) $\frac{9}{8R}$

(4) $\frac{8R}{9}$

33)

Find out following ratio for photon

$(\nu_{max.})_{Lyman} : (\nu_{max})_{Brackett}$

(1) 1 : 16

(2) 16 : 1

(3) 4 : 1

(4) 1 : 4

34) The ratio of minimum wavelengths of Lyman & Balmer series will be :–

(1) 1.25

(2) 0.25

(3) 5

(4) 10

35) The first emission line of Balmer series for H spectrum has the wave no. equal to ?

(1) $\frac{9R}{40}$

(2) $\frac{7R}{144}$

(3) $\frac{3R}{4}$

(4) $\frac{5R}{36}$

36) If the shortest wavelength (λ_{min}) of H-atom in Lyman series is x, then longest wavelength (λ_{max}) in Balmer series of He^+ is :-

(1) $\frac{9x}{5}$

(2) $\frac{36x}{5}$

(3) $\frac{x}{4}$

(4) $\frac{5x}{9}$

37)

The frequency of radiation emitted when the electron falls from n = 4 to n = 1 in a H-atom will be :

(Given R = 1.1×10^7 m^{-1}, C = 3×10^8 m/sec) :-

(1) 3.09×10^{15} s^{-1}

(2) 2.00×10^{15} s^{-1}

(3) 1.54×10^{15} s^{-1}

(4) 1.03×10^{15} s^{-1}

38) Which series have highest energy in hydrogen spectrum :-

(1) Balmer

(2) Brackett

(3) Pfund

(4) Lyman

39)

Among all the following particles moving with same speed, which one have maximum de-broglie's wavelength

(1) alpha particle

(2) electron

(3) proton

(4) all have same de-broglie's wavelength

40) The de broglie wavelength of a 66 kg man sking down kufri hill in shimla at 1×10^3 m/sec is:-

(1) 1×10^{-36} m

(2) 1×10^{-37} m

(3) 1×10^{-38}m

(4) 1×10^{-39} m

41)

If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is 4Å, the circumference of the orbit will be :-

(1) 4Å

(2) 4 nm

(3) 16 Å

(4) 16 nm

42) de-Broglie equation is-

(1) $h = \frac{\lambda}{mv}$

(2) $E = h\nu$

(3) $E = mc^2$

(4) $\lambda = \frac{h}{mv}$

43) No. of wave in fourth orbit :.

(1) 4

(2) 5

(3) 0

(4) 1

44) Match the following Column-I to Column-II.

	Column-I		Column-II
(A)	Radius of nth shell in bohr model of atom	(P)	$\propto \frac{Z^2}{n^3}$
(B)	Angular momentum of electron	(Q)	$\propto -\frac{Z^2}{n^2}$
(C)	Energy of electron in n^{th} shell	(R)	$\propto \frac{n^2}{Z}$
(D)	Frequency of electron in n^{th} shell	(S)	$\propto n$

(1) (A) – (S); (B) – (R); (C) – (Q); (D) – (P)

(2) (A) – (S); (B) – (P); (C) – (R); (D) – (Q)

(3) (A) – (R); (B) – (S); (C) – (Q); (D) – (P)

(4) (A) – (R); (B) – (P); (C) – (Q); (D) – (S)

45) Splitting of spectral lines under influence of magnetic field is called :-

(1) Zeeman effect

(2) Stark effect

(3) Photoelectric effect

(4) Polarisation

BIOLOGY

1) Generally marine and mostly asymmetrical animals are :

(1) Coelenterates like hydra
(2) Sponges like leucosolenia
(3) Poriferans like Euspongia
(4) Helminth like tapeworm

2) Undifferentiated layer mesoglea is found in :

(1) Dentalium
(2) *Hyla*
(3) *Obelia*
(4) Euspongia

3) Which of the following animal has bilateral symmetry in larval stage and radial symmetry in adult stage ?

(1) Star fish
(2) *Balanoglossus*
(3) *Salpa*
(4) *Doliolum*

4) Which one of the following characteristics is not shared by coelenterates and ctenophores ?

(1) Diploblastic
(2) Radial symmetry
(3) Tissue grade body organisation
(4) Ciliary comb plates

5) Urochordate animals have :-

(1) Notochord that extends from head to tail region
(2) Notochord is present throughout larval stages and adult life
(3) Notochord present only in adult stages
(4) Notochord present only in larval stages

6) Alternation of generation between asexual and sexual phases of an organism is known as :-

(1) Metastasis
(2) Metachrosis
(3) Metagenesis
(4) Metamorphosis

7) Tube with in tube body plan (digestive system) is absent in :

(1) *Taenia*

(2) *Ascaris*

(3) Earthworms

(4) Cockroach

8) **Statement - I :** In open circulation blood is pumped out of heart and the cell and tissues are directly bathed in it.
Statement - II : In close circulation blood is circulated through series of vessles of varying diameters.

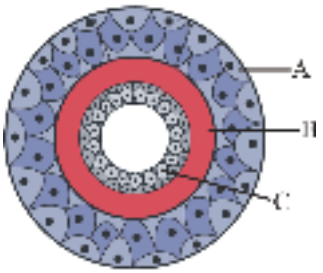
(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is incorrect and statement-II is correct.

(4) Statement-I is correct and statement-II is incorrect.

9) Identify A, B, C in given diagram :-



T.S. of Diploblastic animal

(1) A – Ectoderm, B – Mesoderm, C–Endoderm

(2) A – Ectoderm, B–Mesoglea, C–Endoderm

(3) A – Mesoderm, B–Ectoderm, C–Endoderm

(4) A – Ectoderm, B–Endoderm, C–Mesoglea

10) Which is not a sponges?

(1) Sycon

(2) Spongilla

(3) Scypha

(4) Sea anemone

11) Animals having dorso-ventrally flattened body are :-

(1) Coelenterates

(2) Ctenophores

(3) Platyhelminthes

(4) Aschelminthes

12) Locomotion by 8 ciliary comb plates found in :

(1) *Aurelia*

(2) *Ctenoplana*

(3) *Taenia*

(4) *Nereis*

13) Out of following in which phylum only sexual reproduction occurs ?

(1) Protozoa
(2) Porifera
(3) Coelenterata
(4) Ctenophora

14) Which of the following statements about certain given animals correct ?

(1) Flatworms are true coelomates
(2) Round worms are Pseudocoelomates
(3) Molluscs are acoelomates
(4) Insects are Psuedocoelomates

15) Phylum of animals having single opening to the outside that serves both as mouth and anus is :-

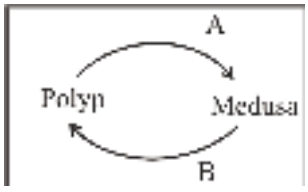
(1) Platyhelminthes
(2) Mollusca
(3) Echinodermata
(4) Hemichordata

16) Pick the odd pair:-

(1) Cellular level - Porifera
(2) Tissue level - Aschelminthes
(3) Organ level - Platyhelminthes
(4) Organ system level - Annelida

17) Solenocytes or flame cells present in :-

(1) Ascaris
(2) Wuchereria
(3) Taenia solium
(4) Trichinella

18) 

Identify A, B in the given figure :-

	A	B
(1)	Sexual reproduction	Asexual reproduction

(2)	Asexual reproduction	Sexual reproduction
(3)	Asexual reproduction	Asexual reproduction
(4)	Sexual reproduction	Sexual reproduction

(1) 1
(2) 2
(3) 3
(4) 4

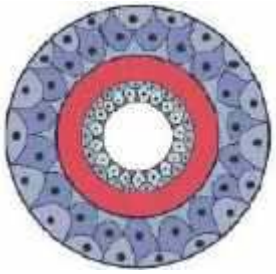
19) Spongilla commonly known as _____ :-

(1) Boring sponge
(2) Fresh water sponge
(3) Bath sponge
(4) Jelly fish

20) Pore through which water moves out of a sponge is called as ?

(1) Ostia
(2) Osculum
(3) Ossicle
(4) Osteon

21) Given diagram below is the body organisation of animal, select the correct phylum which animal



belongs :-

(1) Platyhelminthes
(2) Ctenophora
(3) Aschelminthes
(4) Annelida

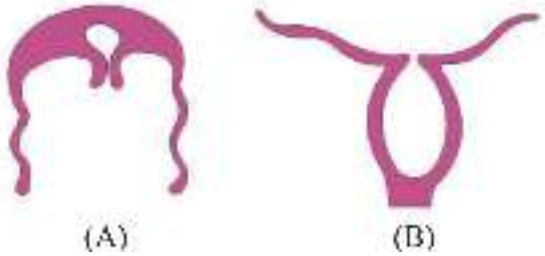
22) "Brain coral" is :

(1) *Meandrina*
(2) *Hydra*
(3) *Physalia*
(4) *Obelia*

23) The types of symmetry in given diagram found in – 

(1) Arthropods
(2) Platyhelminthes
(3) Ctenophores
(4) Annelids

24) Which statement is not true for the given forms A and B?



(1) Structure A is medusa
(2) Structure B is polyp
(3) Structure B is mainly found in jelly fishes
(4) Structure B is sessile form

25) When the body can be divided into identical left and right halves in only one plane. It is called as ?

(1) Radial symmetry
(2) Asymmetry
(3) Bilateral symmetry
(4) Biradial symmetry

26) Identify the feature which is applicable to most members of the phylum represented by the organism shown in the figure below :-



(1) Pseudocoelomate with triploblastic condition

(2) Parapodia for locomotion
(3) Fertilization internal and flame cells for excretion
(4) Mostly ectoparasite with anterio-posteriorly flattened body

27) Which of the following statements about connective tissues is incorrect?

(1) Areolar tissue acts as packing tissue between organs.
(2) Tendons connect muscles to bones.
(3) Ligaments are less elastic than tendons.
(4) Cartilage has cells embedded in a firm matrix.

28) Cilia are most commonly found in epithelial lining of:

(1) Stomach
(2) Small intestine
(3) Trachea
(4) Urinary bladder

29) **Statement I:** Squamous epithelium is thin and flat.
Statement II: It is found in alveoli of lungs.

(1) Both statements I & II are true.
(2) Both statements I & II are false.
(3) Statement I is true, Statement II is false.
(4) Statement I is false, Statement II is true.

30) Choose incorrect :

(1) Tight junctions prevent leakage.
(2) Desmosomes provide strength.
(3) Gap junctions prevent diffusion.
(4) Adhering junctions hold cells together.

31) Which type of connective tissue contains fibers and fibroblasts that are compactly packed and show a regular pattern of orientation?

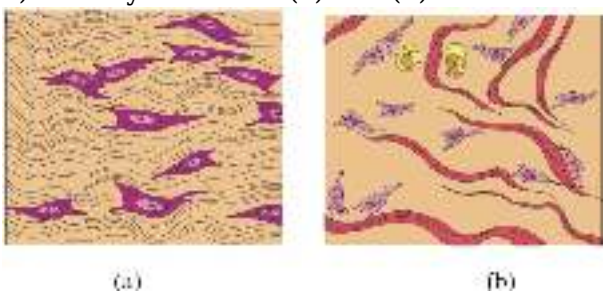
(1) Dense irregular connective tissue
(2) Areolar tissue
(3) Dense regular connective tissue
(4) Adipose tissue

32) Maximum intercellular space is present in ?

(1) Areolar tissue
(2) Adipose tissue
(3) Bone

(4) Cartilage

33) Identify the tissue (a) and (b) shown below and select the right option with example -



Options :

	Tissue	Example	Function
(1)	(b) Special connective tissue	Cartilage	Locomotion
(2)	(a) Dense irregular connective tissue	Tendon	Attach Skeletal Muscle to bone
(3)	(b) Dense Regular	Bone	Support
(4)	(a) Dense Regular Connective Tissue	Ligament	Connect Bone to Bone

(1) 1

(2) 2

(3) 3

(4) 4

34) Blood, bones and cartilages are the examples of :-

(1) Loose connective tissue

(2) Dense regular connective tissue

(3) Dense irregular connective tissue

(4) Specialised connective tissue

35) Which tissue is a reservoir of stored energy in the form of lipid ?

(1) Blood

(2) Muscle

(3) Adipose tissue

(4) Liver

36) Which of the following is not secreted by exocrine glands ?

(1) milk

(2) Earwax

(3) Earwax, digestive enzymes and oils

(4) Hormones

37) Epithelium specific for diffusion and filtration is:-

(1) squamous epithelium
(2) cuboidal epithelium
(3) columnar epithelium
(4) columnar epithelium with cilia

38) Wrong about connective tissue is ?

(1) Their special function is linking and supporting
(2) They range from soft to specilised type
(3) In all connective tissues, the cells secretes fibres
(4) cartilage and bones are example of connective tissue

39) Cuboidal epithelium is commonly found in

(1) lining of stomach
(2) Tubular parts of nephron
(3) skin
(4) inner surface of hollow organs

40) True about simple epithelium is :-

(1) Composed of two/more cell layers
(2) Functions as a lining for body cavities
(3) Has protective function
(4) Cells are loosely packed

41) Fallopian tube having which type of epithelium ?

(1) Simple squamous
(2) Simple cuboidal
(3) Stratified epithelium
(4) Ciliated epithelium

42) Out of following how many are the examples of simple columnar epithelium :
DCT of nephron, Acini of pancreas, Liver, Iris of eye, Bowman's capsule, Inner lining of blood vessel, Intestine

(1) Two
(2) Four
(3) Five
(4) Six

43) In which type of connective tissue cells and fibres are loosely arranged ?

(1) Dense connective
(2) Skeletal connective
(3) Fluid connective
(4) Loose connective

44) Non-nucleated and dead cells with Keratin protein found in which structure ?

(1) Vagina
(2) Pharynx
(3) Oesophagus
(4) Epidermis of skin

45) Epithelium consist of two or more cell layers and has protective function is :-

(1) The epithelium covering the moist surface of buccal cavity
(2) Found in tubular part of nephron
(3) Found in linning of stomach
(4) Mainly present in the inner surface of fallopian tubes

46) Choose the correct match :-

	I	II
(1)	Robert Brown	Nucleus
(2)	Schleiden & Schwann	Cytoplasm
(3)	Singer and Nicolson	Cell wall
(4)	George Palade	Lysosome

(1) 1
(2) 2
(3) 3
(4) 4

47) *Omnis cellula e cellula* was given by:

(1) Rudolf Virchow
(2) Schwann
(3) Schleiden
(4) Robert Brown

48) An animal cell can not be differentiated from plants cell in having:

(1) Cell wall
(2) Central vacuole
(3) Centrosome

(4) Ribosomes

49) RER is frequently observed in cell which are actively involved in ……A…..and are extensive and continuous with the outer membrane of …B…….. What are A and B?

(1) A: Lipid synthesis; B: Nucleus
(2) A: Protein synthesis; B: Nucleus
(3) A: Protein synthesis; B: Mitochondria
(4) A: Lipid synthesis; B: Mitochondria

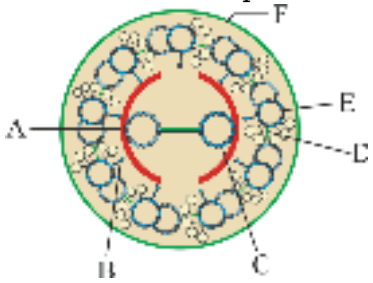
50) Which of the following is important for osmoregulation in *Amoeba*?

(1) Food vacuole
(2) Digestive vacuole
(3) Sap vacuole
(4) Contractile vacuole

51) The inner compartment of mitochondria is filled with homogenous substance called as :

(1) Matrix
(2) Stroma
(3) Cytosol
(4) PMS

52) In a given diagram of cilia which letter represents Central sheath, Interdoublet bridge, and Radial spoke respectively ?



(1) A, B and D
(2) B, F, D
(3) A, D and B
(4) D, A, E

53) Nuclear DNA exists as a complex of proteins called_____that condenses into____during_____

(1) Chromatids, chromosomes, cell division
(2) Chromosomes, chromatin, interphase
(3) Chromatin, chromosome, interphase
(4) Chromatin, chromosome, cell division

54) (I) The shape of the cells may vary with the function they perform
(II) Human RBC is about 7.0 µm in diameter

(III) Cytoplasm is the main arena of cellular activities
(IV) Various chemical reactions occur in cytoplasm to keep the cell in the living state
Which of the following statement is correct ?

(1) All are correct
(2) Only I and II are correct
(3) Only IV is correct
(4) All are incorrect

55) Chromatin contains

(1) DNA
(2) non histone proteins
(3) histones, RNA
(4) all of these

56) The fluid nature of membrane is also important from which of the following function

(1) cell growth, endocytosis
(2) formation of intercellular junctions
(3) secretion, cell division
(4) all of these

57) Which cell organelles takes part in the formation of lysosomes :-

(1) Endoplasmic reticulum
(2) Golgi bodies
(3) Both 1and 2
(4) Mitochondria

58) Which one of the following is not a constituent of cell membrane ?

(1) Phospholipids
(2) Cholesterol
(3) Glycolipids
(4) Pectin

59) Which of following is not common in chloroplasts & mitochondria ?

(1) Both are present in animal cells
(2) Both contain their own genetic material
(3) Both are present in eukaryotic cells
(4) Both are present in plant cells

60) Arm ratio in metacentric condition is :-

(1) 1 : 1

(2) 1 : 2

(3) 1 : 3

(4) 2 : 3

61) Microtubules are the constituents of :-

(1) Cilia, Flagella and Peroxisomes

(2) Spindle fibres, Centrioles and Cilia

(3) Centrioles, Spindle fibres and Chromatin

(4) Centrosome, Nucleosome and Centrioles

62) Match the column-I and II select the correct option:-

	Column-I		**Column-II**
a	Cisternae	I	Proteinaceous structure in cytoplasm
b	Thylakoid	II	Infolding of inner mitochondrial membrane
c	Cristae	III	Flattened membranous sac
d	Microtubule	IV	Membranous structure in Golgi complex

(1) a-IV, b-III, c-II, d-I

(2) a-II, b-IV, c-III, d-I

(3) a-I, b-II, c-IV, d-III

(4) a-IV, b-III, c-I, d-II

63) In fluid mosaic model of plasma membrane :-

(1) Upper layer is non-polar and hydrophilic

(2) Polar layer is hydrophobic

(3) Phospholipids form a bimolecular layer in middle part

(4) Proteins form a middle layer

64) Maximum cell organelles are synthesized in which phase of cell cycle ?

(1) G_1

(2) S

(3) G_2

(4) Prophase

65) If the cell had 2n number of chromosomes at G_1, after S phase the number of chromosome will be-

(1) 2n

(2) 4n

(3) n

(4) 8n

66) Meiosis involves :-

(1) Formation of two diploid daugther cells at the end of process.
(2) Pairing of homologous chromosomes and recombination between them.
(3) Two cycles of DNA replication.
(4) Two sequential cycles of cell division but single cycle of nuclear division.

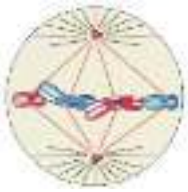
67) If the number of chromosomes in a diploid cell is 24, then find out the number of chromosomes during anaphase :-

(1) 24
(2) 12
(3) 48
(4) 36

68) Choose the **correct** option for the stage at which morphology of chromosome is easily studied :-

(1) 

(2) 

(3) 

(4) 

69) Which one of the following events is incorrect for cell cycle ?

(1) Tubulin protein synthesis occurs in G_2-phase
(2) Maximum cell growth occurs in M-phase
(3) DNA synthesis occurs only during one specific stage
(4) Centriole duplication occurs in S-phase

70) Crossing over takes place in :-

(1) Zygotene
(2) Pachytene
(3) Diplotene
(4) Diakinesis

71) Nuclear envelope reappears at :-

(1) Metaphase
(2) Prophase
(3) Anaphase
(4) Telophase

72) In which stage of mitosis, the chromosomes are composed of two chromatids ?

(1) Prophase & metaphase
(2) Anaphase and telophase
(3) Prophase and telophase
(4) Metaphase and anaphase

73) During which phase(s) of cell cycle, amount of DNA in a cell remains at 4C level if the initial amount is denoted as 2C ?

(1) G_0 and G_1
(2) G_1 and S
(3) Only G_2
(4) G_2 and M

74) How many chromosomes shall be present in a diploid cell at mitotic anaphase if its egg cell has ten chromosomes ?

(1) 10
(2) 20
(3) 30
(4) 40

75) Read the following four statements (A – D) regarding meiosis - I :-
(A) Condensation of Chromatin
(B) Separation of sister chromatids
(C) Segregation of homologous Chromosomes
(D) Crossing over between non-Sister Chromatids.
How many of the above events occur during meiosis-I ?

(1) Four
(2) Three

(3) Two

(4) One

76) Cytokinesis in animal cell and plant cell takes place by :-

(1) Animal cell - Cell plate - Centripetal
Plant cell - Furrowing - Centrifugal

(2) Animal cell - Furrowing - Centrifugal
Plant cell - Cell plate - Centripetal

(3) Animal cell - Furrowing - Centripetal Plant cell - Cell plate - Centrifugal

(4) Animal cell - Furrowing - Centrifugal Plant cell - cell plate - centrifugal

77) If in a cell 20 bivalent is present in prophase-I then how many chromosome will be there in each daughter cell after meiosis-II :-

(1) 10

(2) 20

(3) 40

(4) 5

78) Mark incorrect statements :-
(A) Meiosis involves only a single cycle of DNA replication
(B) Four haploid cells are formed at the end of meiosis–II
(C) Mitosis restricted to diploid cells only
(D) In yeast, cell cycle takes about 90 minutes.

(1) A and B

(2) A and C

(3) Only C

(4) All are correct

79) Which is the correct order of cell cycle ?

(1) $G_1 \rightarrow S \rightarrow G_2 \rightarrow$ Prophase $\rightarrow$ Metaphase $\rightarrow$ Anaphase $\rightarrow$ Telophase $\rightarrow$ cytokinesis

(2) $G_2 \rightarrow G_1 \rightarrow S \rightarrow$ P–M $\rightarrow$ A $\rightarrow$ T $\rightarrow$ Cytokinesis

(3) $S \rightarrow G_1 \rightarrow G_2 \rightarrow$ P–M $\rightarrow$ A $\rightarrow$ T $\rightarrow$ Cytokinesis

(4) Prophase $\rightarrow$ Metaphase $\rightarrow$ Anaphase $\rightarrow$ Telophase $\rightarrow$ Cytokinesis $\rightarrow S \rightarrow G_2 \rightarrow G_1$

80) Number of chromatids in each chromosome at anaphase is-

(1) One in mitosis, one in meiosis-I and two in meiosis-I

(2) One in mitosis, two in meiosis-I and one in meiosis-II

(3) Two in mitosis, one in meiosis-I and two in meiosis-II

(4) Two in mitosis, two in meiosis-I and two in meiosis-II

81) **Assertion :** Meiosis produces four genetically dissimilar cells.

Reason : Crossing over or gene exchange takes place in meiosis.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

82) If the n=16 in plant cell then what is possible in metaphase - I of meiosis ?

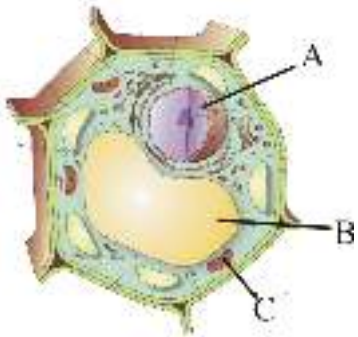
(1) 32 Bivalents

(2) 16 Tetravalents

(3) 16 Bivalents

(4) 64 Bivalents

83) The given figure represents a plant cell. identify the labelled parts, match them with functions (i –iii) and select the correct option :-



(i) Aerobic respiration
(ii) Storage of excretory product.
(iii) Stores the cell's hereditary material.

(1) A – i, B – ii, C – iii

(2) A – ii, B – i, C – iii

(3) A – iii, B – ii, C – i

(4) A – i, B – iii, C – ii

84) How many statement are wrong in following :-
(a) Centrioles are necessary for the division of all eukaryotic cells
(b) Nacked DNA found in prokaryotes because of absence of nuclear membrane
(c) Cell plate formation occurs during divisions of plant cells
(d) RER involved in lipid synthesis

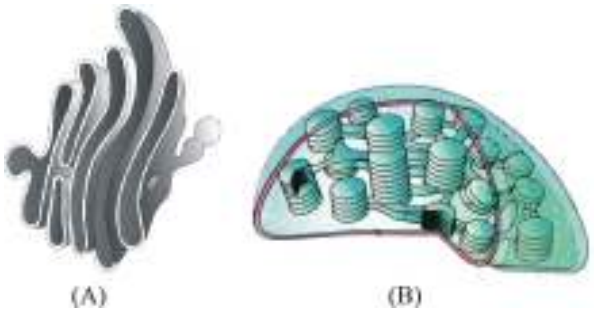
(1) Three

(2) Two

(3) One

(4) Four

85) The two sketches (A and B) given below, represents two different type of cell organelles which one of these correctly identified in the option given, along with its correct location and function?



		Cell-organelles	Location	Function
(1)	(B)	Chloroplast	All eukaryotic cell	Photosynthesis
(2)	(A)	Golgi body	Prokaryotic cell	Secretion
(3)	(B)	Leucoplast	Plant cell	Photosynthesis
(4)	(A)	Golgi body	Eukaryotic cell	Packaging of material

(1) 1
(2) 2
(3) 3
(4) 4

86) Which of the following represents the features of lysosomes ?

(1) A lower pH than cytoplasm
(2) Presence of acid hydrolases
(3) Presence of water soluble pigments
(4) Both (1) and (2)

87) Which of the following is true for flagella ?

(1) Oar like movement
(2) Compartively longer than cillia
(3) Comparatively shorter than cillia
(4) Number per cell is very high

88) A person observed a cell under a microscope. He found that the chromosomes had moved to the spindle equator and were aligned along the metaphase plate. Spindle fibres were attached to the kinetochores of the chromosomes and connected to both poles. In which stage of cell division is the cell?

(1) Prophase
(2) Metaphase
(3) Anaphase
(4) Telophase

89) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The M phase strarts with the nuclear division, corresponding to the separation of daughter chromosomes (cytokinesis) and usually ends with division of cytoplasm (karyokinesis).
Reason (R) : The duration of cell cycle can vary from organism to organism and also from cell type to cell type.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false

(4) Assertion is false but Reason is true

90) Given below are two statements:
Statement I : At the first stage of karyokinesis, the chromosomes that have reached their respective poles decondense and lose their individuality. And Nucleolus, golgi complex and ER reform.
Statement II: In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	1	1	2	3	3	2	2	2	1	3	1	2	4	1	1	2	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	3	3	3	1	2	3	4	1	2	3	4	3	2	1	4	4	1	3
Q.	41	42	43	44	45															
A.	2	3	1	1	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	2	4	2	2	2	1	4	1	2	2	3	3	3	2	1	2	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	2	4	3	1	4	1	1	2	3	1	2	2	4	1	1	4	2	3
Q.	86	87	88	89	90															
A.	3	4	1	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	1	4	4	3	1	1	2	4	3	2	4	2	1	2	3	2	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	3	3	3	3	3	3	1	3	3	1	4	4	3	4	1	3	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	4	4	1	1	1	4	2	4	1	3	4	1	4	4	3	4	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	1	3	1	1	2	3	3	2	2	4	1	4	4	2	3	2	3	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	3	1	4	4	2	2	4	4										

SOLUTIONS

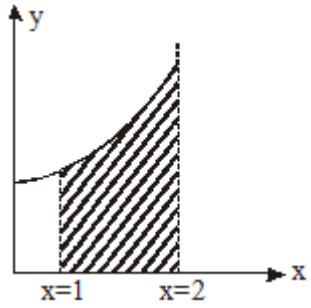
PHYSICS

1) $\int_0^2 x^{-1/2}dx = \frac{x^{1/2}}{1/2}$

$= 2\left[x^{1/2}\right]_0^{\infty}$

$= 2[\infty - 0] = \infty$

2) **Question Explanation :** A y-x curve is shown below :



If, $y = 3x^2 + 1$

Then, from x=1 to x=2; what will be the area (A) under the curve = ?

Concept : Application of Integration : Area Under Curve

Formula : Area under y-x curve $\Rightarrow A = \int_{x_1}^{x_2} y \cdot dx$

Solution :

Area under the curve is given

$A = \int_{x_1}^{x_2} y \cdot dx$

$A = \int_1^2 (3x^2 + 1) \cdot dx$

$A = \int_1^2 3x^2 \cdot dx + \int_1^2 1 \cdot dx$

$A = \left[\frac{3x^3}{3}\right]_1^2 + [x]_1^2 \qquad \left\{\because \int_{x_1}^{x_2} x^n.dx = \left[\frac{x^{n+1}}{n+1}\right]_{x_1}^{x_2}\right\}$

$A = [(2)^3 - (1)^3] + [2 - 1]$

$A = (8 - 1) + (1)$

$A = 8$ units

Final Answer : option (1)

$$\int_0^1 (3x^2 - 4x + 1)\,dx = \left[3\frac{x^3}{3} - \frac{4x^2}{2} + x\right]_0^1$$

$$= 1 - \frac{4}{2} + 1 = 0$$

3)

4) **Question Explanation :** In this question, we have to solve $\int (\sin x + \cos x)\,.dx$.

Concept : Indefinite Integration

Solution:

$\int (\sin x + \cos x)\,.dx = \int \sin x.dx + \int \cos x + c$ = –cosx + sinx + c

Final Answer: option(1)

5)

$R = \sqrt{5^2 + 5^2 + 2 \times 5 \times 5 \cos 60^\circ}$

$R = 5\sqrt{3}$ N

$$\int_{\pi/6}^{\pi/2} \sin x \, dx = [-\cos x]_{\pi/6}^{\pi/2}$$

$$= \left[-\cos\frac{\pi}{2}\right] - \left[-\cos\frac{\pi}{6}\right] = 0 + \frac{\sqrt{3}}{2}$$

6)

7) **Question Explanation :** In this question, we have to evaluate the $\int [4\cos(t) + t^2]\,dt = ?$

Concept : Indefinite integration of trigonometric functions and algebraic functions.

Solution :

$$\int [4\cos(t) + t^2]\,dt = 4\sin(t) + \left(\frac{t^{2+1}}{2+1}\right) + C = 4\sin(t) + \frac{t^3}{3} + C$$

Final Answer : Option (3)

8)

Explain Question : Evaluate the integral ⇒ $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 dx = ?$

Concept : This question is based on the indefinite integration of the algebric functions, the inverse algebric functions & the constant values.

Solution :

$$\int \left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2 dx = \int \left[(\sqrt{x})^2+\left(\frac{1}{\sqrt{x}}\right)^2+2(\sqrt{x})\left(\frac{1}{\sqrt{x}}\right)\right]dx$$

$$= \int \left[x+\frac{1}{x}+2\right]dx$$

$$= \frac{x^2}{2}+\ln(x)+2x+C$$

$$= \frac{x^2}{2}+\log_e(x)+2x+C$$

Final answer : 2

9) **Explain question :** Evaluate the integral $\int_{R_1}^{R_2} \frac{GMm}{r^2}dr,$ representing gravitational work from R_1 to R_2.

Concept : This question is based on $\int \frac{1}{r^2}dr = -\frac{1}{r}$

Solution : $\int_{R_1}^{R_2} \frac{GMm}{r^2}dr = GMm\int_{R_1}^{R_2} \frac{1}{r^2}dr$

$$= GMm\left[-\frac{1}{r}\right]_{R_1}^{R_2} = GMm\left(-\frac{1}{R_2}+\frac{1}{R_1}\right)$$

$$= GMm\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$ **Final answer :** 2

10) **Question Explanation:** In this question, we have to determine the sum of the provided infinite G.P.

Concept: Sum of an infinite geometric progression

Solution:

$1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+...\infty$

$a = 1$ (First term)

$r = \frac{1}{3}$ (common ratio)

Sum of an infinite geometric progression is given by

$$S\infty = \frac{a}{1-r}$$

$$S\infty = \frac{1}{1-1/3}$$

$$S\infty = \frac{1}{2/3}$$

$$S\infty = \frac{3}{2}$$

Final answer: Option (2)

11) **Explaining :** Find sum of ∞ G.P. series.

Concept : This question is based on $S_\infty = \frac{a}{1-r}$

Solution :

$a = \frac{1}{5}$

$r = \frac{1}{5}$

$S_\infty = \frac{\frac{1}{5}}{1-\frac{1}{5}} = \frac{\frac{1}{5}}{\frac{4}{5}}$

$S_\infty = \frac{1}{4}$

Final Answer : (1)

12) **Explain Question:**

Identify the correct graph representation for the linear equation $y = \sqrt{3}x$

Concept:

Slope-intercept form of a straight line y mx + c, where m = tan(θ).

Solution:

$y = \sqrt{3}x - 4$

$m = \sqrt{3}$

c = – 4

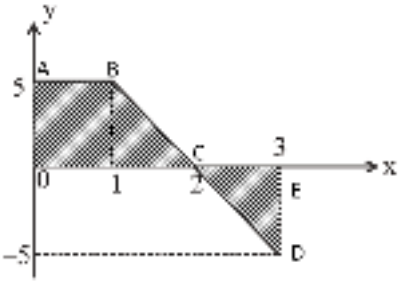
$\tan(\theta) = \sqrt{3}$

θ = 60°

Final Answer: Option (3)

13)

Explain Question :



For the shown plot, find $\Rightarrow \int_0^3 y\,dx$ = ?

Concept : This question is based on definite integration within given limits is equal to the area under curne.

Formula : $\int_{x=a}^{x=b} ydx$ = Area under "y-x" curve from x = a to x = b.

Solution :

$\int_{x=a}^{x=b} ydx$ = Area of trapezium OABC + Area of triangle CDE

$= + \left[\frac{1}{2} \times (OC + AB) \times OA\right] + \left[\frac{1}{2} \times CE \times ED\right]$

$$= +\left[\frac{1}{2} \times (2+1) \times (5)\right] + \left[\frac{1}{2} \times (1) \times (-5)\right]$$
$$= +\frac{1}{2} \times 3 \times 5 - \frac{1}{2} \times 1 \times 5$$
$$= +\frac{15}{2} - \frac{5}{2}$$
$$= +\frac{10}{2}$$
$$= +5$$

Final answer : 1

14)

Explain the Question : We need to find the total area under the y–x curve from x = 0 to x = 4, considering positive and negative contributions.

Concept : This question is based on area under curve = sum of areas of geometric regions above x-axis (positive) and below x-axis (negative).

Solution :

From x = 0 to x = 1 : triangle with base = 1, height = 2

$A_1 = \frac{1}{2} \times 1 \times 2 = 1$

From x = 1 to x = 2 : rectangle with width = 1, height = 2

$A_2 = 1 \times 2 = 2]$

From x = 2 to x = 4 : rectangle with width = 2, height = –3 (below x-axis)

$A_3 = 2 \times (-2) = -6$

Total area = $A_1 + A_2 + A_3 = 1 + 2 - 6 = -3$

Final answer : 2

15) $$\int_0^1 9x^8 dx + \int_0^{\pi/2} \cos x \, dx$$
$$= \left[9 \times \frac{x^9}{9}\right]_0^1 + [\sin x]_0^{\pi/2}$$
$$= [1 - 0] + [1 - 0] = 2$$

16) **Question Explanation :**

We have to find average value of acceleration a = 2 (t–1) from t = 0 to t = 2 Sec.

Concept :

This question is based on Average value of a function.

Solution :

$$a_{avg} = \frac{\int_a^b a \, dt}{\int_a^b dt}$$

$$a_{avg} = \frac{\int_0^2 2(t-1)\,dt}{\int_0^2 dt} \Rightarrow \frac{\int_0^2 (2t-2)\,dt}{\int_0^2 dt}$$

$$\Rightarrow \frac{\left[2\frac{t^2}{2} - 2t\right]_0^2}{[t]_0^2} \rightarrow \frac{\left[2^2 - 2\times 2\right]}{2} = 0$$

Final Answer : Option (1)

17) **Question Explanation :** In this question, we have to find the displacement with the help of given (v – t) graph from t = 0 to t = 3 sec.

Concept: Area under velocity-time graph.

Solution: Area under velocity-time graph (with sign) gives the displacement.

Area under (v – t) graph from t = 0 to t = 3sec

Area under (v – t) graph = Area of trapezium + area of rectangle

Area $= \frac{1}{2}(1+2)(10) + (1\times -6)$

Area = 15 – 6 = 9 m

Final Answer: option(1)

18) **Question Explanation :** In this question, we have to find area of shaded region for the given graph.

Concept: Application of integration

Solution:
Area under the curve represented by a function f(x) within the range $[x_1, x_2]$ is given by

$$\int_{x_1}^{x_2} f(x)dx$$

 Area of shaded region $\Rightarrow \int_{x_1}^{x_2} f(x)dx = \int_{-1}^{1} x^3 dx$

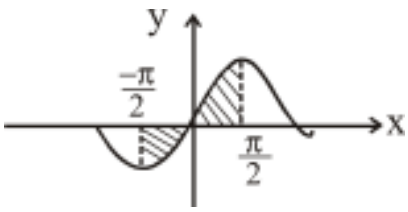
$$= \frac{1}{4}\left[x^4\right]_{-1}^{1}$$

$$= \frac{1}{4}\left[(1)^4 - (-1)^4\right] = 0$$

Final Answer: option(2)

19) $\left[e^x\right]_{-1}^{1} = e - \frac{1}{e} = \frac{e^2-1}{e}$

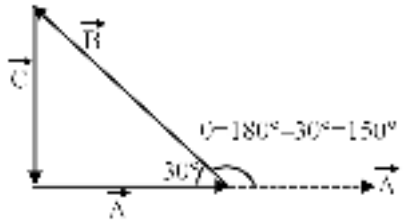
20) $A = \int_0^6 x^2 dx = \left[\frac{x^3}{3}\right]_0^6 = \frac{216}{3} = 72$



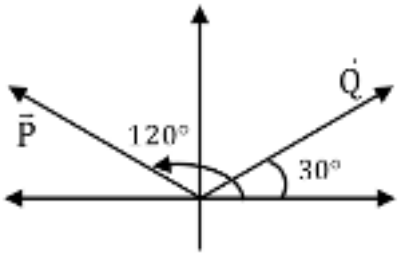
21)
Area = 0

22) $A = \int_1^4 (2x-3)dx = 6\text{units}$

23) **Explain the Question :** In this question we have to find the angle between the vectors $\vec{A}$ and $\vec{B}$.
Concept : This question is based on angle between two vectors.
Solution : Angle between vectors is angle formed by either joining their heads or tails (≤180°) in the given figure.



On parallely shifting the vector $\vec{A}$, we can see that the angle between the vectors $\vec{A}$ and $\vec{B}$.
θ = 180° – 30° = 150°
Final Answer : 3



24)
So, $\vec{P} \perp \vec{Q}$
$\vec{P} + \vec{Q} = \sqrt{P^2 + Q^2}$

25) **Calculation :**

(i) θ = 90° $\quad R = 2F\cos\left(\frac{90}{2}\right) \Rightarrow \sqrt{2}f$

(ii) θ = 120° $\quad R = 2F\cos\left(\frac{120}{2}\right) \Rightarrow F$

(iii) $\theta = 60°$ $R = 2F\cos\left(\frac{60}{2}\right) \Rightarrow \sqrt{3}F$

Answer :

Magnitude of resultant force is maximum when $\theta = 60°$ so Answer will be option (3)

26) **Question Explanation:**

We have to calculate resultant of $\vec{A}$ and $\vec{B}$.

Concept:

Resultant of two vectors

The magnitude of the resultant R is given by:

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

Solution:

Given:

A. $|\vec{A}| = 2$

B. $|\vec{B}| = \sqrt{3}$

C. $\theta = 30^\circ$

Using formula:

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$= \sqrt{2^2 + (\sqrt{3})^2 + 2(2)(\sqrt{3})\cos 30^\circ}$$

$$= \sqrt{4 + 3 + 4\sqrt{3} \cdot \frac{\sqrt{3}}{2}}$$

$$= \sqrt{4 + 3 + 4 \cdot \frac{3}{2}}$$

$$= \sqrt{7 + 6}$$

$$= \sqrt{13}$$

Final Answer : Option (1)

27) $\vec{a} = 0.5\hat{i} + 0.8\hat{j} + c\hat{k}$

$|\vec{a}| = 1 = \sqrt{(0.5)^2 + (0.8)^2 + c^2}$
$\Rightarrow c^2 = 1 - [0.25 + 0.64]$
$\Rightarrow c = \sqrt{1-(0.89)} = \sqrt{0.11}$

28) If initial and final position is same result will be zero
Using triangle law
$\vec{B} + \vec{C} - \vec{A} = \vec{0}$

29) When $|\vec{A}| = |\vec{B}|$; $R = 2A\cos\frac{\theta}{2}$
$\Rightarrow A = 2A\cos\frac{\theta}{2} \quad \Rightarrow 1 = 2\cos\frac{\theta}{2}$
$\Rightarrow \frac{1}{2} = \cos\frac{\theta}{2} \quad \Rightarrow 60° = \frac{\theta}{2}$
$\Rightarrow \theta = 120°$

30) **Question Explanation :** In this question, we have to find the unit vector $\hat{A}$ in the direction of a vector $\vec{A}$.
Concept : Unit vector
Solution :
Unit vector for a given vector $\vec{A}$ is given by

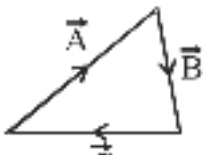
$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

Unit vector = vector divided by its magnitude.
• Ensures $|\hat{A}| = 1$
• Points in the same direction as $\vec{A}$
Final answer : Option (1)

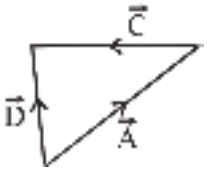
31)

Explaining the Question :- Find the angle between vectors $\vec{A}$ and $\vec{B}$ when $|\vec{A}|$=12, $|\vec{B}|$=
5, =13, and $\vec{C} = \vec{A} + \vec{B}$.
Concept:- This question is based on Magnitude of resultant : $|\vec{C}|^2 = |\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta$
Solution :-
$|\vec{C}|^2 = |\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta$
$13^2 = 12^2 + 5^2 + 2 .12.5 \cos\theta$
$169 = 144 + 25\ 120 \cos\theta \Rightarrow 120 \cos\theta = 0 \Rightarrow \theta = 90°$
Final Answer :- 2

32) $15 - 10 \le R \le 15 + 10$
$5 \le R \le 25$
2N is not possible

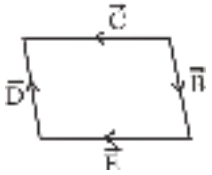
33) (i) 

$\vec{A}+\vec{B}+\vec{E}=0$

(ii) 

$\vec{A}+\vec{C}=\vec{D}$

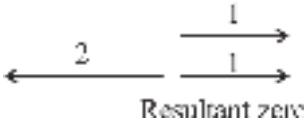
$\vec{C}-\vec{D}=-\vec{A}$

(iii) 

$\vec{C}=\vec{B}+\vec{E}+\vec{D}$

$-\vec{D}=\vec{B}+\vec{E}-\vec{C}$

34) $A-B \le R \le A+B$



35) Resultant of three forces can be zero if magnitude of third force must present between maximum & minimum value of resultant of two forces.

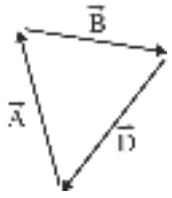
36)

Given: $F_1 = F_2 = F$, $R = 40\sqrt{3}N$ and $\theta = 60°$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$\Rightarrow 40\sqrt{3} = \sqrt{F^2 + F^2 + 2F^2\cos 60} = \sqrt{3F^2}$$

$$\Rightarrow \quad F = 40N$$

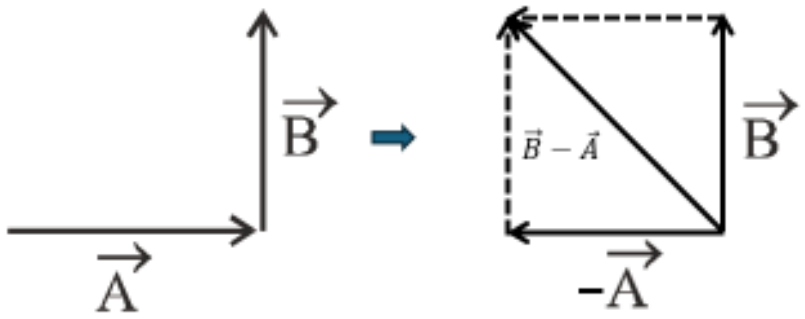
37) 

$\vec{A}+\vec{B}+\vec{D}=0 \vec{D}=-(\vec{A}+\vec{B})$

38) **Explanation Question :** Question asks diagram of $\vec{B}-\vec{A}$ is.

Concept: Subtraction of vectors.

Solution:



Final answer: Option (4)

39) **Question Explanation:** To find the angle between $\vec{A}$ and $\vec{B}$ under given condition.

Solution:- Let $|\vec{A} - \vec{B}| = |\vec{A}| = |\vec{B}| = a$

Concept:- Subtraction of vectors.

Formula:- $R' = |\vec{A} - \vec{B}| = 2a\sin\frac{\theta}{2}$

$R' = 2a\sin\frac{\theta}{2}$

$a = 2a\sin\frac{\theta}{2}$

$\frac{1}{2} = \sin\frac{\theta}{2} \Rightarrow \sin\frac{\theta}{2} = \sin 30$

$\boxed{\theta = 60^\circ}$

40) $|\hat{A} - \hat{B}| = 2 \times 1 \sin \theta/2 \qquad = 2 \sin\frac{\theta}{2}$

41) **Explain Question:**

We have to calculate angle between $\vec{a}$ & $\vec{b}$

Concept:

Subtraction of vectors

Solution:

$2a \sin \theta/2 = 1 \quad a = 1$

$\sin \theta/2 = 1/2$

$\theta/2 = 30°$

$\theta = 60°$

Final answer: (2)

42) $R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

$2F = \sqrt{4F^2 + F^2 + 4F^2\cos\theta}$

$4F^2 = 5F^2 + 4F^2 \cos \theta$

$\cos\theta = -\frac{1}{4} \Rightarrow \theta = \left(-\frac{1}{4}\right) \cos^{-1}$

43) $F_1 + F_2 = 10$

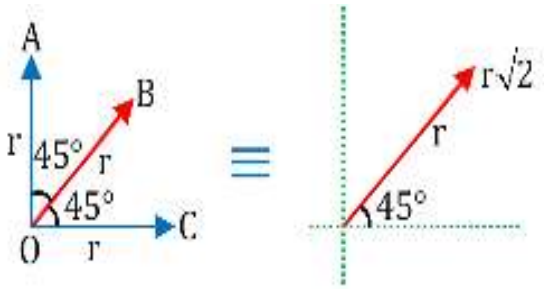
$F_1 - F_2 = 6 \Rightarrow F_1 = 8, \quad F_2 = 2$

New $F_1' = 8 + 3 = 11$

$F_2' = 2 + 3 = 5$

$R = \sqrt{11^2 + 5^2 + 2 \times 11 \times 5\cos 90^\circ}$

$= \sqrt{121 + 25} = \sqrt{146}$



44)

$= r + r\sqrt{2}$

$= r\left(1 + \sqrt{2}\right)$

45) Let the common ratio be x

$|\vec{A}| = 3x, |\vec{B}| = 5x$

$|\vec{A} + \vec{B}| = \sqrt{(3x)^2 + (5x)^2 + 2(3x)(5x)\cos 60^\circ}$

or, $35 = \sqrt{9x^2 + 25x^2 + 15x^2}$

or, $35 = 7x$

$\Rightarrow x = 5$

$|\vec{A}| = 15 \; ; \; |\vec{B}| = 25$

CHEMISTRY

46)

n = 3, z = 3

$r_3 = 0.529 \times \frac{3^2}{3} = 0.529 \times \frac{9}{3} = 0.529 \times 3 = 1.587 \text{ Å}$

47) $\frac{r_2}{r_5} = \frac{(2)^2}{3} \times \frac{3}{(5)^2}$

48) $\frac{r_{2,He^+}}{r_{3,Li^{2+}}} = \frac{a_0 \times \frac{2^2}{2}}{a_0 \times \frac{3^2}{3}} = \frac{2}{3}$

$\Rightarrow r_{3,Li^{2+}} = r_{2,He^+} \times \frac{3}{2} = 105.8 \times \frac{3}{2} = 158.7pm$

49) $\frac{r_1}{r_2} = \frac{4}{1} = \frac{n_1^2}{1} \times \frac{1}{n^2} \Rightarrow \frac{n_1^2}{n_2^2} \times \frac{4}{1}$

$\frac{n_1}{n_2} = \frac{2(L)}{1(K)} = \frac{4(N)}{2(L)}$

50) $(v_2)_{He^+} = 2.188 \times 10^6 \times \frac{2}{2} = 2.18 \times 10^6 m/sec$

51)

$v \propto \frac{z}{n}$

$\frac{v_H}{v_{Li^{+2}}} = \frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$

52) $T \propto \frac{n^3}{z^2}$ z = 1

$\frac{T_1}{T_2} = \frac{n_1^3}{n_2^3}$ $n_1 = 1$, $n_2 = 2$

$\frac{T_1}{T_2} = \frac{1}{8}$

53)

Frequency relation : $f \propto \frac{Z^2}{n^3}$

for He^+ (third orbit state → n = 3, Z = 2)

$$f_{He^+} \propto \frac{2^2}{3^3} = \frac{4}{27}$$

For H (second orbit → n = 2, Z = 1)

$$f_H \propto \frac{1^2}{2^3} = \frac{1}{8}$$

$$\frac{f_{He^+}}{f_H} = \frac{4/27}{1/8} = \frac{32}{27}$$

Final answer : 1

54) $\Delta E = |E_3 - E_2| = 1.89$

55)

When e^- jumps from
Higher → Lower
Energy is released.

56)

Second shell (n = 2):

$E_2 = -13.6 \times \frac{(2^2)}{(2^2)} = -13.6 \times \frac{4}{4} = -13.6$ eV.

Final Answer : Option (2)

57)

$V = 2.18 \times 10^6 \, Z/n$ m/sec

$$f = \frac{v}{2\pi r} \Rightarrow f \propto \frac{Z^2}{n^3}$$

$$r = 0.529 \frac{n^2}{Z} \, \text{Å}$$

$$E = -13.6 \frac{Z^2}{n^2} \text{eV/atom}$$

58)

3rd excuted state ⇒ $n_2 = 4$; $n_1 = 1$

$$\Delta E = 13.6 \times z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\Delta E = 13.6 \times (2)^2 \left[\frac{1}{1^2} - \frac{1}{4^2}\right]$$

$$= 13.6 \times 4 \times \left[\frac{1}{1} - \frac{1}{16}\right]$$

$$= 13.6 \times 4 \times \frac{15}{16} = 51 \text{ eV}$$

59) On increasing radius both P.E and T.E increases as P.E or T.E $\alpha -\frac{1}{r}$

60)

λ_{min} ⇒ Last line of Paschen series

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = R \times 3^2 \left[\frac{1}{3^2} - \frac{1}{\infty^2}\right]$$

$$\Rightarrow R \text{ or } \lambda = \frac{1}{R}$$

61)

P.E. = –3.02 eV
T.E. = –1.51 eV
n = 3 E.S. = 2

62)

$(E_2 - E_1) = 10.2\ Z^2$
$(E_3 - E_2) = 1.89\ Z^2$
$(E_4 - E_3) = 0.66\ Z^2$

Clearly:
$(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$

Final Answer : option (1)

63) n = 3, TE = 1.51 eV
PE = 2 × TE
= 2 × (–1.51)
= –3.02 eV

64)

$E_1 = -13.6$ eV, $E_2 = -3.4$ eV, $E_3 = -1.51$ eV, $E_4 = -0.85$ eV

- P: $E_2 - E_1 = -3.4 - (-13.6) = 10.2$ eV → (iii)
- Q: $E_3 - E_2 = -1.51 - (-3.4) = 1.89$ eV → (i)
- R: $E_4 - E_2 = -0.85 - (-3.4) = 2.55$ eV → (iv)
- S: $E_3 - E_1 = -1.51 - (-13.6) = 12.09$ eV → (ii)

P-(iii), Q-(i), R-(iv), S-(ii)

Final Answer → Option 3

65)

First separation energy = $E_\infty - E_2$
for Be^{+3} ion = $[0 - (-3.4)]_H \times Z^2$
= 3.4 × 16 = 54.4 eV

66)

$r \propto n^2$, $V \propto \frac{1}{n}$

67) = $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

$$= \frac{(4-2)(4-2+1)}{2}$$
$$= \frac{2 \times 3}{2} = 3$$

68)

The transition is to a final energy level of $n_f = 2$, which corresponds to the Balmer series.
Final Answer (2)

69)

Energy release :
Higher orbit → Lower orbit
$\lambda_{max} \Rightarrow E_{min}$
n = 1 to n = 2 (absorb)
n = 4 to n = 1 (release) $E_4 - E_1 = 12.75$
n = 2 to n = 3 (absorb)
n = 2 to n = 1 (release) $E_2 - E_1 = 10.2$

70)

$\gamma_{max} \Rightarrow E_{max}$
1 to ∞, (absorb)
2 to 1, $E_2 - E_1 = 10.2$ eV
3 to 1, $E_3 - E_1 = 12.09$ eV
3 to 2, $E_3 - E_2 = 1.89$ eV

71)

Transitions only produce line UV (3 → 1 and 2 → 1) and visible (3 → 2) regions
No Infrared region

72)

Lyman series ⇒ UV region

73)

I line of Balmer series : 3 → 2
II line of Balmer series : 4 → 2
III line of Balmer series : 5 → 2

74)

Lyman	Balmer	Paschen	Brackett	Pfund
UV	visible	IR	IR	IR

↓	↓	↓	↓	↓
3	2	1	0	0

Total 1 IR line

75)

No. of lines in Paschen series = $n_2 - 3$
$= 8 - 3$
$= 5$ lines

76) **Assertion :** When e^- lands on n = 2 ⇒ Balmer (Visible region)
Reason :- Lyman series radiation belongs to UV region.

77)

Last line of Paschen series : ∞ → 3

$$\frac{1}{\lambda} = R \times (1)^2 \times \left(\frac{1}{3^2} - \frac{1}{\infty^2}\right)$$

$$\Rightarrow \frac{1}{\lambda} = \frac{R}{9} \Rightarrow \lambda = \frac{9}{R}$$

78)

$\gamma_{max} \Rightarrow E_{max}$ = Las line
$(\gamma_{max})_{lyman} \Rightarrow \infty \rightarrow 1$
$(\gamma_{max})_{Brackett} \Rightarrow \infty \rightarrow 4$

$$\frac{(\gamma_{max})_{lyman}}{(\gamma_{max})_{Brackett}} = \frac{CRZ^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)}{CRZ^2\left(\frac{1}{4^2} - \frac{1}{\infty^2}\right)} = \frac{16}{1}$$

79)

Lyman → $n_2 = \infty$ to $n_1 = 1$.
Balmer → $n_2 = \infty$ to $n_1 = 2$.

$$\frac{\frac{1}{\lambda_L}}{\frac{1}{\lambda_B}} = \frac{\left(RZ^2\right)\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right]}{\left(RZ^2\right)\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right]}$$

$$= \frac{\lambda_L}{\lambda_B} = \frac{1}{4} = 0.25$$

Final answer : Option 2

80)

I line of Balmer series : 3 → 2

$$\frac{1}{\lambda} = \bar{\nu} = R(1)^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$= R \times \left(\frac{5}{36}\right)$$

81) $$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

For Lyman series $n_1 = 1$, $n_2 = \infty$

(shortest wavelength)

$$\frac{1}{\lambda} = R(1)^2\left(\frac{1}{(1)^2} - \frac{1}{\infty}\right) = \frac{1}{x} = R$$

$$R = \frac{1}{x}$$

For Balmer series $n_1 = 2$, $n_2 = 3$

(longest wavelength)

$$\frac{1}{\lambda} = R(2)^2\left[\frac{1}{(2)^2} - \frac{1}{(3)^2}\right]$$

$$\frac{1}{\lambda} = \frac{1}{x} \times 4\left[\frac{1}{4} - \frac{1}{9}\right] = \frac{1}{x} \times 4 \times \frac{5}{36} \Rightarrow \boxed{\lambda = \frac{9}{5}x}$$

82) $$\gamma = CRZ^2\left(\frac{1}{1^2} - \frac{1}{4^2}\right)$$

$$= 3 \times 10^8 \times 1.1 \times 10^7 \times (1)^2 \times \frac{15}{16}$$

83)

What to find : Series with maximum ΔE.

Energy of transition : $E = 13.6\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ eV

Series :

Lyman ($n_1 = 1$)

Balmer ($n_1 = 2$)

Brackett ($n_1 = 4$)

Pfund ($n_1 = 5$)

Maximum ΔE occurs for lowest $n_1 \rightarrow$ lyman series.

Final answer : 4

84) $$\lambda = \frac{h}{mV}$$

n , V → constant

$$\Rightarrow \lambda \propto \frac{1}{m}$$

$(\lambda_{ab})_{max} \Rightarrow (mass)_{min}$

e^- has minimum mass

85)

$$\lambda = \frac{6.6 \times 10^{-34}}{(66) \times (1 \times 10^{3})} = 1 \times 10^{-38} \text{m}$$

The de-Broglie wavelength is 1×10^{-38}. The correct option is 3.

Final Answer : option (3)

86)

n = 4, λ = 4 A°
Circumference = 2πr = nλ
Circumference = nλ
Circumference = 4 × 4 A°
Circumference = 16 A°

87)

Theoritical.

88) No. of orbit in 4th orbit = 4

89)

Notes

90) **Solution:**
• **Zeeman effect:** Splitting of atomic spectral lines caused by magnetic field interaction with electron energy levels.
• **Stark effect:** Splitting due to electric field, not magnetic.
• **Photoelectric effect:** Emission of electrons by light, unrelated here.
• **Polarisation:** Orientation of light waves, not splitting.

Final Answer: Option (1)

BIOLOGY

91) **Explain Question :** The question asks which animals are generally marine and mostly asymmetrical.
Concept : This question is based on Porifera
Solution: The correct answer is option 3.

A. Euspongia (bath sponge) → Porifera, marine and asymmetrical.
B. Hydra (Coelenterata) → radial symmetry, not asymmetrical.
C. Leucosolenia (sponge) → Porifera, radial symmetrical.

D. Tapeworm (Platyhelminth) → bilaterally symmetrical.

Final Answer: 3.

92) NCERT 50

93) NEERT Page No. # 54

94) **Question Explanation:** Unique trait in ctenophores
Concept: Ciliary comb plates
Solution: Both coelenterates and ctenophores are diploblastic, have radial symmetry, and show tissue-grade organization. Only ctenophores have ciliary comb plates.
Final Answer: Ciliary comb plates

95)

Explain Question:
Identify notochord condition in Urochordata.

Concept:
This question is based on Urochordata (Tunicates).

Solution:
The correct answer is option 4.

A. In Urochordates, notochord is present only in the larval stage (in the tail), but it disappears in the adult.

Final answer: 4

96) NCERT (XI) Pg. # 50

97) **Question Explanation:** Identify the organism that lacks a "tube within tube" body plan.
Concept: Platyhelminthes
Solution: The **Tapeworm** (**Taenia**) lacks a true digestive system and, therefore, the "tube within tube" body plan; food is absorbed through its body surface.
Final Answer: 1

98) **Question Explanation**: The question asks to determine the correctness of the two statements regarding types of circulatory systems — open and closed circulation.

Concept: Circulatory pathways

Solution:

A. Statement I: Correct — describes open circulation accurately.

B. Statement II: Correct — describes closed circulation accurately.

Final Answer: Option – 1 → Both statement-I and statement-II are correct.

99) **Question Explanation:** Identifying germ layers in a diagram.**Concept:** Diploblastic Germ Layers
Solution: A **diploblastic** animal has two main germ layers. The outer layer is the **ectoderm** (A), and the inner layer is the **endoderm** (C). The space between them is filled with a non-cellular, undifferentiated layer called the **mesoglea** (B).
Final Answer: A – Ectoderm, B–Mesoglea, C–Endoderm

100)

Question Explanation: Identify the organism that is not a sponge.
Concept: Porifera, Cnidaria
Solution: Sea anemone (**Adamsia**) is a **Coelenterate** , not a sponge.
Final Answer: 4

101)

NCERT XI Page # 42

102)

Explain Question: This question is asking about which organism uses 8 ciliary comb plates for locomotion.

Concept: This question is based on Phylum Ctenophora.

Solution: Locomotion by 8 ciliary comb plates found in Ctenophores.
Eg. Ctenoplana

Final Answer: Option (2)

103)

Explain Question: Out of following in which phylum only sexual Reproduction occurs.

Concept: This question is based on Phylum ctenophore

Solution: ctenophore have only sexual Reproduction.

Final Answer: (4)

104)

Explaining:
Which of the following statements about certain given animals is correct?

Concept: Animal body cavities are classified as acoelomate (no cavity), pseudocoelomate (false cavity), or eucoelomate (true cavity).
Solution:

Option 1: Flatworms are acoelomates, not true coelomates.

Option 2: Roundworms (Phylum Nematoda) are pseudocoelomates.

Option 3: Molluscs are true coelomates, not acoelomates.

Option 4: Insects (Arthropoda) have hemocoel, not a pseudocoel.

105) The correct answer is **1. Platyhelminthes.**

A. **Platyhelminthes (flatworms):** They have a single opening to their digestive system, called the gastrovascular cavity. This opening serves as both the mouth (for taking in food) and the anus (for expelling waste).

106)

Explain Question: Pick the odd pair?

Concept: This question is based on Animal kingdom.

Solution: Phylum Aschelminths have organ system level of organisation.
Final Answer: (2)

107)

NCERT XI pg no.51

108)

Explain Question : Identify A & B in Obelia life cycle.
Concept : This question is based on Cnidaria
Solution : The correct answer is **option 2** .
In Obelia, **polyp stage reproduces asexually** to give rise to medusae.
Medusae then **reproduce sexually** to form polyps again.
This alternation between asexual (polyp) and sexual (medusa) generations is called **metagenesis**.
Final Answer : 2

109) NCERT (XIth) Pg. # 50

110)

The correct answer is 2. Osculum
Here's why:

A. Ostia are the tiny pores on the surface of a sponge through which water enters.
B. Osculum is the large opening at the top of a sponge through which water exits.

111)

NEW NCERT XI, Page No. # 38

112)

Explanation :

A. Meandrina is commonly known as Brain Coral due to its grooved, maze-like surface resembling a human brain.
B. It belongs to Phylum Cnidaria (Coelenterata) and is a type of stony coral found in marine environments.

Correct Answer: Option (1) Meandrina

113) NCERT Pg # 47

114) NCERT XI, Page # 50

115)

Explain Question: Identify the type of body symmetry where an animal can be divided into two identical halves along a single plane.

Concept: Symmetry - Basis classification of animal

Solution: The correct answer is option 3.

A. Bilateral symmetry is the body plan where the body can be divided into identical left and right halves by only one plane. This is a characteristic of most complex animals.

Final answer : 3

116) NCERT (XI) Pg. # 51, Fig. 4.9 (a)

117) **Solution :** Ligaments are more elastic than tendons due to higher elastic fibre content.
Final Answer : (3)

118) **Solution :** Ciliated epithelium lines trachea to move mucus.
Final Answer : (3)

119) **Solution :** Both statements I & II are true.
Final Answer : (1)

120) **Solution :** Gap junctions allow diffusion.
Final Answer : (3)

121) **Solution :** In tissues like tendons and ligaments, collagen fibers are arranged in parallel bundles to provide high tensile strength.
Final Answer : (3)

122)

Areolar tissue, with its loosely arranged fibers and abundant ground substance, has the largest amount of intercellular space compared to the other options listed. This structural feature allows it to perform functions such as cushioning, support, and diffusion of nutrients and gases.

123)

Explain Question : Labelling of Diagram with its example and function ?

Concept : This question is based on Connective tissue.

Solution : Option (4)
Dense Regular C.T. —— Ligament —— Connect Bone to Bone
Final Answer : option (4)

124)

Question Explanation: The question is asking which category includes blood, bone and cartilage.

Concept: Types of connective tissue.

Solution: Blood, bone and cartilage are all considered specialised connective tissues because they have specific structures and functions different from loose or dense connective tissues.

Final Answer: Option 4

125) NCERT XI, Pg # 120 (Summary)

126)

Question Explain : Which of the following is not secreted by exocrine glands?

Concept: This question is Based on classification of glands(animal tissue).

Solution:
Exocrine glands secrete substances through ducts (e.g., milk, sweat, earwax, digestive enzymes)Hormones (secreted by endocrine, not exocrine glands).

Final answer: 4

127) NCERT Pg # 101/7.1.1

128) NCERT Pg # 103/7.1.2

129) NCERT Pg # 101/7.1.1

130) NCERT Pg # 101/7.1.1

131)

Explain Question : The question is asking about the specific type of epithelium that lines the fallopian tube, which is involved in the transport of the ovum.

Concept : This question is based on types of simple epithelium.

Solution : The fallopian tube is lined with ciliated epithelium. The cilia help in the movement of the ovum from the ovary toward the uterus.

Final Answer : option (4)

132)

Liver, Uterus

133)

Explain Question : The question is asking about the type of connective tissue where both cells and fibers are loosely arranged.

Concept : This question is based on Types of connective tissue.

Solution : Loose connective tissue, such as areolar tissue, has cells and fibers loosely arranged in a semi-fluid matrix, allowing flexibility and support.

Final Answer : option (4)

134)

Correct Answer: Option 4 (Epidermis of skin)

Explanation

A. The epidermis of the skin is made up of stratified squamous keratinized epithelium, where the outermost cells are dead, non-nucleated, and filled with keratin protein.
B. The vagina, pharynx, and oesophagus have stratified squamous non-keratinized epithelium.

135) NCERT Pg.# 101/102

136)

NCERT-XI, Page # 87-88

137) NCERT XI Pg # 126

138) NCERT XI Pg # 129

139) NCERT XI Pg # 133

140) NCERT XI Pg # 134

141) NCERT XI Pg # 135

142) NCERT, Pg # 137

143)

NCERT-XI, Pg. # 101

144)

NCERT-XI, Pg. # 88, 89

145)

NCERT-XI, Pg. # 101

146)

NCERT-XI, Pg. # 94

147)

NCERT-XI, Pg. # 96

148)

NCERT-XI, Pg. # 93

149)

NCERT-XI, Pg. # 97, 98

150)

NCERT-XI, Pg. # 101

151)

NCERT-XI, Pg. # 99

152)

NCERT-XI, Pg. # 95, 97, 98

153)

Explanation

According to the fluid mosaic model, the plasma membrane is composed of:

A. A phospholipid bilayer, where hydrophobic tails face inward and hydrophilic heads face outward.

B. Embedded proteins, which help in transport and signaling.

C. Cholesterol and carbohydrates, which provide stability and communication.

Correct Answer: 3. Phospholipids form a bimolecular layer in the middle part

154)

Question explanation: This question asks during which phase of the cell cycle most cell organelles are synthesized.

Concept : Cell Cycle: Phases And Introduction (G_1, S, G_2)

Explanation : Most organelle synthesis occurs during the G_1 phase.

Final answer : G_1

155)

NCERT-XI, Pg. # 121

156)

NCERT-XI, Pg. # 125, 126

157)

Question Explanation: Determining chromosome number during anaphase.

Concept: Mitosis - Anaphase Stage.

Solution:
In anaphase of mitosis, the sister chromatids of each chromosome are pulled apart to opposite poles. Each chromatid is considered an individual chromosome at this point.

If the diploid number of chromosomes is 24 (i.e., 2n = 24), the cell starts with 24 chromosomes. During anaphase, the chromatids separate, so the total number of chromosomes temporarily doubles to 48 (24 × 2).

Final Answer: 3

158) NCERT (XI) Pg. # 165, 166

159)

NCERT-XI, Pg. # 121

160)

NCERT-XI, Pg. # 126

161)

NCERT-XI Pg. # 124

162)

NCERT-XI, Pg. # 127

163)

NCERT-XI, Pg. # 121

164)

NCERT-XI, Pg. # 123

165)

NCERT-XI, Pg. # 126

166)

NCERT-XI, Pg. # 124

167)

NCERT-XI, Pg. # 127

168)

NCERT-XI, Pg. # 122

169) NCERT-XI, Pg. # 121

170)

NCERT-XI, Pg. # 123, 127, 128

171)

Question Explanation
The question is asking whether meiosis produces genetically dissimilar cells and if crossing over explains this outcome.

Concept
significance of meiosis

Solution
Meiosis produces four haploid cells that are genetically different from each other due to mechanisms like crossing over and independent assortment. Crossing over (gene exchange) directly contributes to this genetic dissimilarity.

Final answer: Option 1

172) **Explanation**

A. n = 16 indicates the haploid number of chromosomes, so the diploid number of chromosomes in the plant is 2n = 32.

B. During Metaphase-I, homologous chromosomes pair up to form 16 bivalents (since each bivalent consists of two chromosomes).

Therefore, the correct number of bivalents in Metaphase-I is 16.

Correct Answer: 3. 16 Bivalents

173)

NCERT Pg. # 92

174)

NCERT-XI, Pg. # 95, 100

175)

NCERT-XI, Pg. # 95

176)

NCERT-XI, Pg. # 96

177) NCERT Pg. # 137

178) NCERT-XI, Pg. # 123

179) NCERT, Pg. # 121

180) NCERT-XI, Pg. # 124